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Q1) A Describe the different types of Distributed Denial of Service (DDoS) flooding attacks..

Q1) A) Describe the Different Types of Distributed Denial of Service (DDoS) Flooding Attacks. (8 Marks)

1. Definition

A Distributed Denial of Service (DDoS) attack is a cyberattack in which multiple compromised computers or devices send a large amount of traffic to a target server or network to make its services unavailable to legitimate users.

2. Types of DDoS Flooding Attacks

1. SYN Flood Attack

- The attacker sends a large number of TCP SYN requests to a server.
- The server responds and waits for the connection to complete.
- Many incomplete connections consume server resources.
- Genuine users may be unable to connect.

2. UDP Flood Attack

- The attacker sends a large number of UDP packets to the target.
- The server uses resources to process these packets.
- Network bandwidth and system resources may become overloaded.
- Legitimate users experience service disruption.

3. ICMP Flood Attack

- The attacker sends a large number of ICMP packets, such as ping requests.
- The target spends resources processing or responding to the packets.
- Network bandwidth may be consumed.
- The service may become slow or unavailable.

4. HTTP Flood Attack

- The attacker sends a large number of HTTP requests to a website.
- The server attempts to process every request.
- CPU and memory resources become overloaded.
- The website becomes slow or unavailable.

5. DNS Amplification Attack

- The attacker misuses publicly accessible DNS servers to send large responses toward a victim.
- The victim receives a much larger volume of traffic than the attacker directly sends.
- Network bandwidth becomes overloaded.
- Legitimate network communication is disrupted.

6. TCP Flood Attack

- The attacker generates excessive TCP traffic toward the target.
- The target must process and manage the incoming packets and connections.
- Network and server resources become exhausted.
- Genuine users may lose access to the service.

7. Volumetric Flooding Attack

- Multiple compromised devices send huge amounts of traffic to the victim.
- The traffic consumes the available network bandwidth.
- Legitimate data cannot reach the target properly.
- The network service becomes unavailable.

4. Prevention of DDoS Attacks

- Use firewalls and intrusion prevention systems.
- Apply traffic filtering and rate limiting.
- Use DDoS protection services.
- Monitor unusual network traffic.
- Configure servers to handle sudden increases in traffic.

Conclusion

DDoS flooding attacks attempt to make online services unavailable by overwhelming network bandwidth or system resources. Common types include SYN, UDP, ICMP, HTTP, DNS amplification, TCP, and volumetric flooding attacks. Proper filtering, monitoring, and DDoS protection help reduce their impact.

Q2) B Write a Python program to detect spam emails using predefined keywords

```
email = input("Enter email message: ")

keywords = ["win", "free", "prize", "offer", "lottery"]

email = email.lower()

spam = False
```

```
for word in keywords:
    if word in email:
        spam = True
if spam:
    print("Spam Email")
else:
    print("Not a Spam Email")
```

OR

Q2) B Write a program to display ASCII values of characters before encryption.

```
text = input("Enter a message: ")
print("Characters and ASCII values:")
for ch in text:
    print(ch, "=", ord(ch))
```